Holard –Total soil water content
Chresard –Water available to plants
Echard – Water not available to plants

6.2.3 Topographic factors

The surface features of earth are called **topography**. Topographic influence on the climate of any area is determined by the interaction of solar radiation, temperature, humidity, rainfall, latitude and altitude. It affects the vegetation through climatic variations in small areas (micro climate) and even changes the soil conditions. Topographic factors include latitude, altitude, direction of mountain, steepness of mountain etc.

a. Latitudes and altitudes

Latitudes represent distance from the equator. Temperature values are maximum at the equator and decrease gradually towards poles. Different types of vegetation occur from equator to poles which are illustrated below.

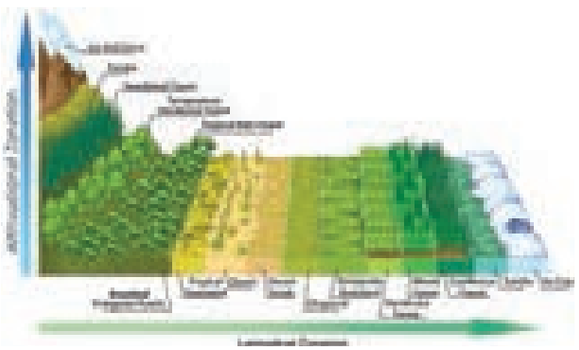


Figure 6.8: Latitudinal and Altitudinal Vegetation

Height above the sea level forms the **altitude**. At high altitudes, the velocity of wind remains high, temperature and air pressure decrease while humidity and intensity of light increases. Due to these factors, vegetation at different altitudes varies, showing distinct zonation.

b. Direction of Mountain

North and south faces of mountain or hill possess different types of flora and fauna because they differ in their humidity, rainfall, light intensity, light duration and temperature regions.

Ecotone - The transition zone between two ecosystems. Example: The border between forest and grassland.

Edge effect – Spices found in ecotone areas are unique due to the effect of the two habitats. This is called edge effect. Example: Owl in the ecotone area between forest and grassland.

The two faces of the mountain or hill receive different amount of solar radiation, wind action and rain. Of these two faces, the windward region possesses good vegetation due to heavy rains and the leeward region possesses poor vegetation due to rain shadows (rain deficit).

Similarly in the soil of aquatic bodies like ponds the center and edge possess different depth of water due to soil slope and different wave actions in the water body. Therefore, different parts of the same area may possess different species of organisms.

c. Steepness of the mountain

The steepness of the mountain or hill allows the rain to run off. As a result the loss of water causes water deficit and quick erosion of the top soil resulting in **poor vegetation**. On the other hand, the plains and valley are **rich in vegetation** due to the slow drain of surface water and better retention of water in the soil.

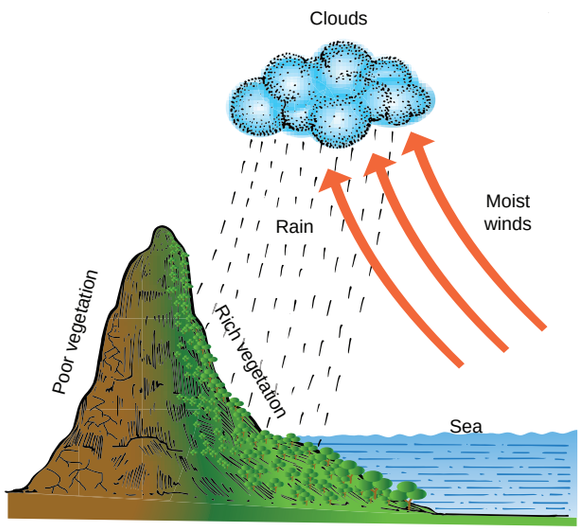


Figure 6.9: Steepness of mountain

TN_GOVT_BIOLOGY-BOTANY_XII_PAGES_122-147 CH 06.indd 117 3/14/2020 11:38:37 AM